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Flag 0 — Create Your Feature Branch

Objective: Create a new feature branch before adding interactions.

Create:

feature/student-interactions

Screenshot:

```
WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ git branch
  feature/data-ui
  feature/exam
  feature/exam_2
* feature/student-list
  feature/widgets
  main

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ git checkout -b feature/student-list
fatal: a branch named 'feature/student-list' already exists

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-list)
$ git checkout -b feature/student-interactions
Switched to a new branch 'feature/student-interactions'

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-interactions)
$ git push -u origin feature/student-interactions
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/student-interactions' on GitHub by visiting:
remote:   https://github.com/mykylicaballero/IT314AB_Caballero/pull/new/feature/student-interactions
remote:
To https://github.com/mykylicaballero/IT314AB_Caballero.git
 * [new branch]      feature/student-interactions -> feature/student-interactions
branch 'feature/student-interactions' set up to track 'origin/feature/student-interactions'.

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-interactions)
$
```

Flag 1 — Wake Up the Buttons

Add at least two buttons inside each Student Card.

Example actions:

- Favorite
- Edit

Objective: Make your Student Card respond to button presses using `onPressed`.

Right now your buttons are only decorations.

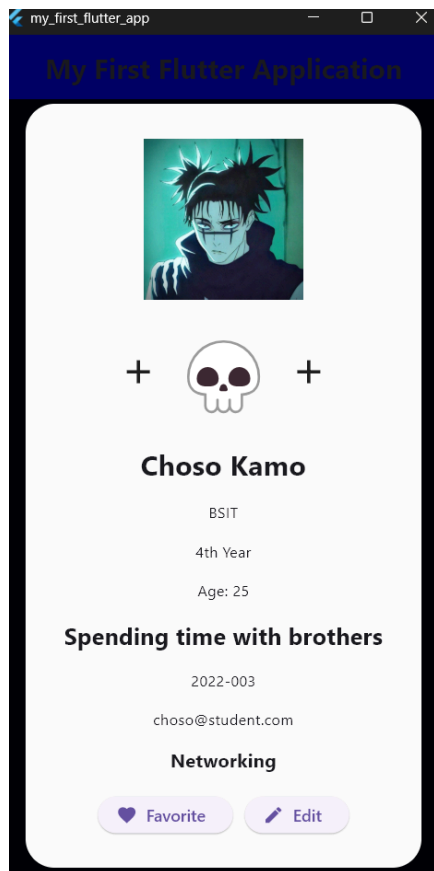
Research how Flutter's `onPressed` works.

The buttons do not need to perform their final behavior yet. They simply need to respond when pressed.

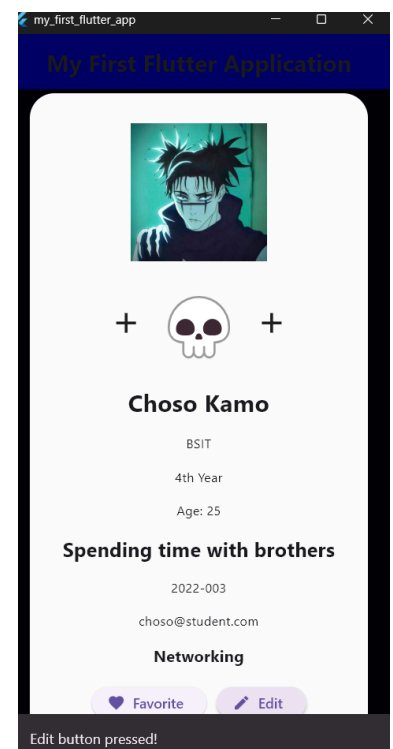
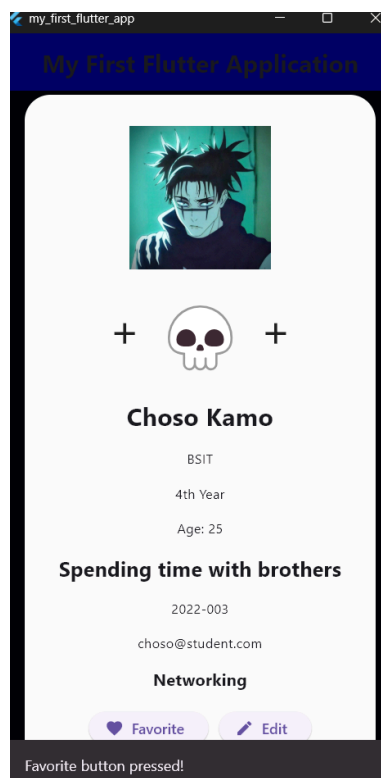
Test

When a button is pressed, your app should visibly react in some way (such as printing a message or showing a temporary notification).

Screenshot:



```
ElevatedButton.icon(  
  onPressed: () {  
    ScaffoldMessenger.of(context).showSnackBar(  
      const SnackBar(  
        content: Text(  
          "Favorite button pressed!",  
        ), // Text  
      ), // SnackBar  
    );  
  },  
  icon: const Icon(Icons.favorite),  
  label: const Text("Favorite"),  
) // ElevatedButton.icon  
const SizedBox(width: 10),  
ElevatedButton.icon(  
  onPressed: () {  
    ScaffoldMessenger.of(context).showSnackBar(  
      const SnackBar(  
        content: Text(  
          "Edit button pressed!",  
        ), // Text  
      ), // SnackBar  
    );  
  },  
  icon: const Icon(Icons.edit),  
  label: const Text("Edit"),  
) // ElevatedButton.icon
```



🚩 Flag 2 — Touch Anywhere

Objective: Make the entire Student Card tappable using `onTap`.

Buttons aren't the only interactive elements.

Research Flutter's `onTap`.

Wrap your Student Card so tapping anywhere on the card triggers an interaction.

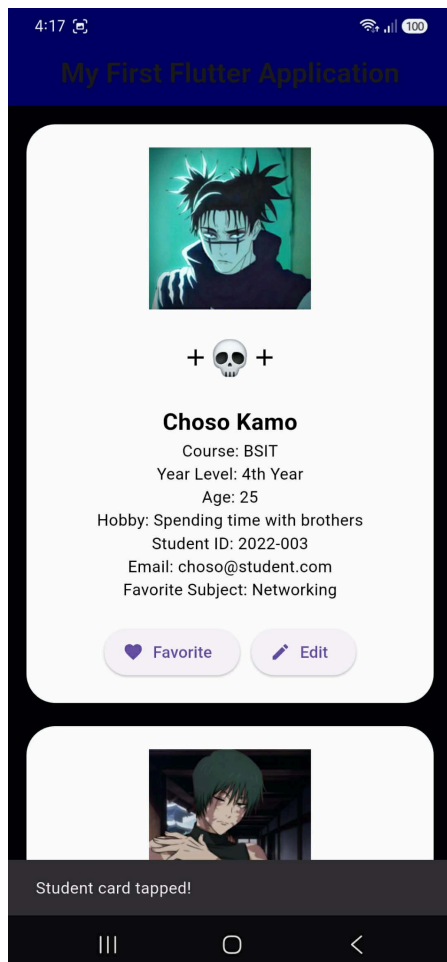
Test

The user should be able to:

- Tap the Favorite button
- Tap the Edit button
- Tap anywhere else on the Student Card

Each interaction should be recognized separately. (One function for the 2 buttons, another for the Student Card)

Screenshot:



```
return GestureDetector(  
  onTap: () {  
    showMessage(  
      context,  
      "Student card tapped!",  
    );  
  },  
);
```

```
ElevatedButton.icon(  
  onPressed: () {  
    showMessage(  
      context,  
      "Favorite button tapped!",  
    );  
  },  
  icon: Icons.favorite,  
);
```

🚩 Flag 3 — Discover `setState`

Objective: Update the screen immediately using `setState`.

So far your app reacts. But does the UI actually change?

Research Flutter's `setState`.

Use it so pressing a button immediately updates something visible on the screen.

Examples include:

- Changing text
- Changing an icon
- Updating a counter

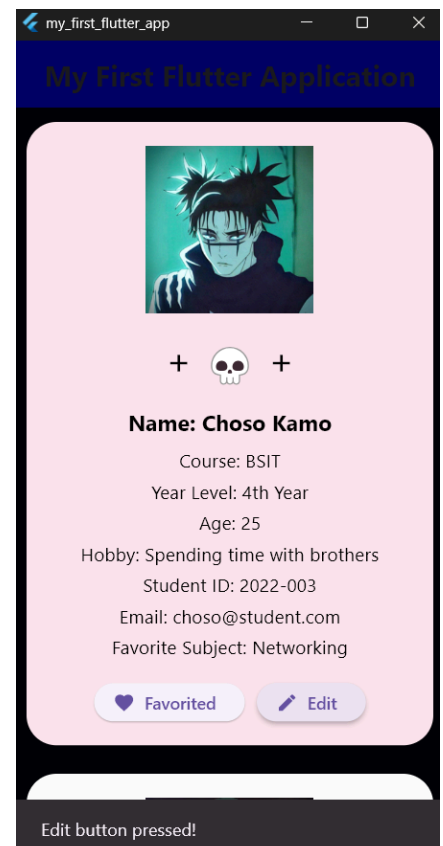
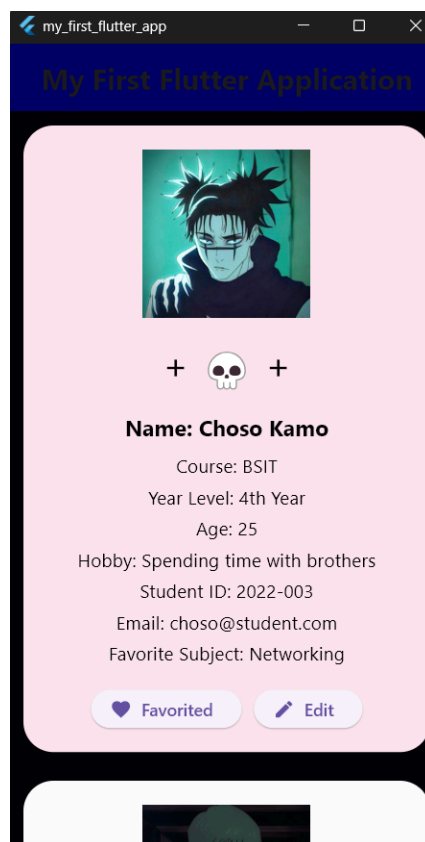
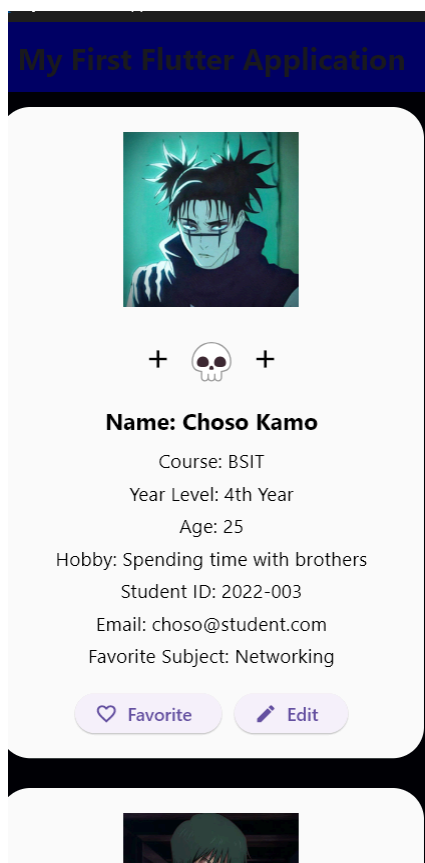
The important part is understanding that `setState` tells Flutter to rebuild the UI.

Test

The screen should visibly change immediately after pressing the button.

Screenshot:

```
class MyApp extends StatefulWidget {  
  // ...  
  setState(() {  
    // ...  
  })  
}
```



🚩 Flag 4 — Favorite Students

Objective: Let administrators mark students as favorites.

Schools often need to highlight important records.

Add a favorite feature that changes a student's appearance when marked.

Possible visual changes include:

- Filled heart/star
- Different card color
- Favorite label

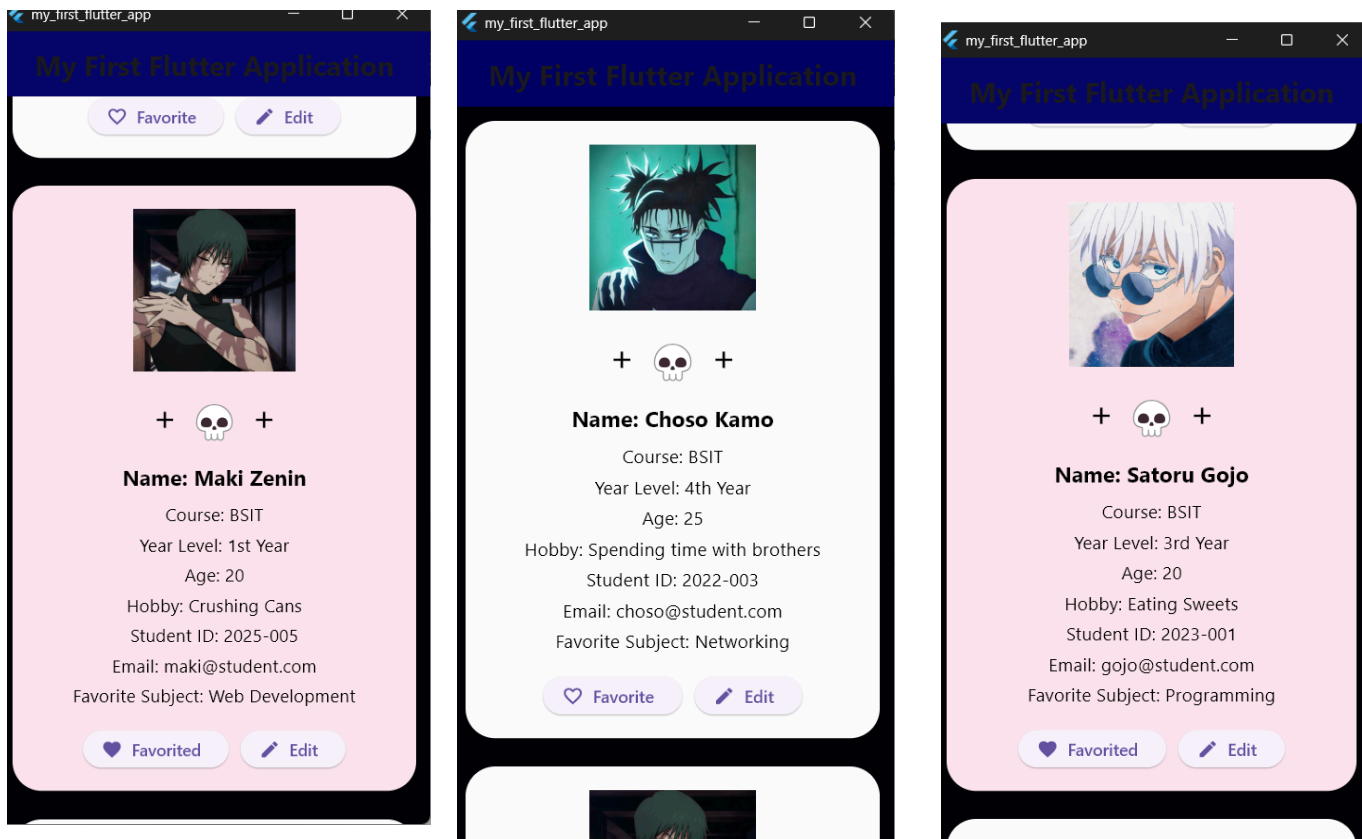
Each student's favorite status should work independently.

Test

- Favorite one student.
- Other students should remain unchanged.
- Favorite another student.

Both should keep their own state.

Screenshot



🚩 Flag 5 — Remove a Student

Objective: Delete a student from the directory.

Research how to remove an item from a Dart [List](#).

Add a Delete action that removes a student from your list.

The UI should immediately update after deletion.

Test

Before:

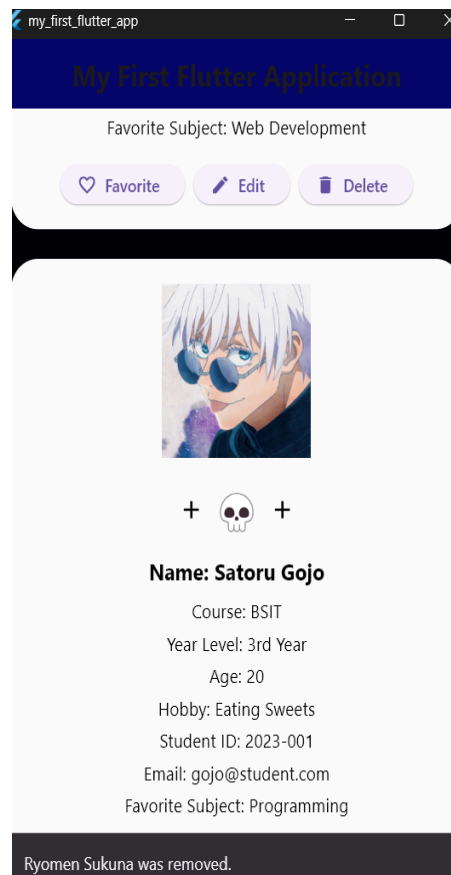
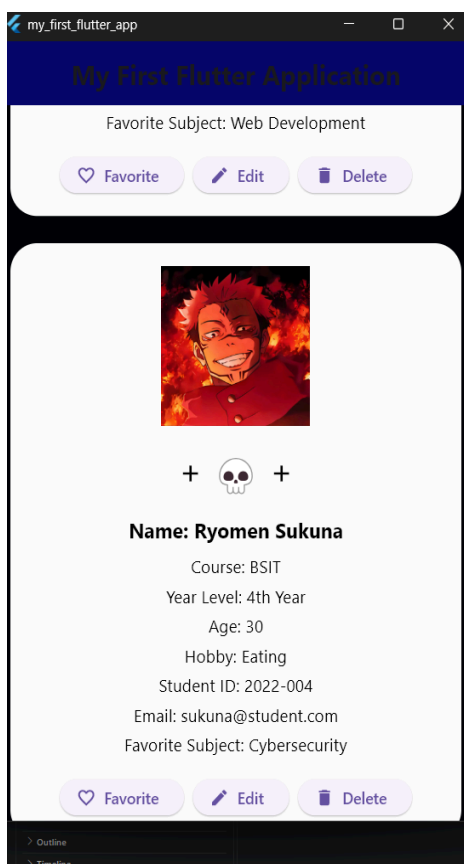
- Student A
- Student B
- Student C

After deleting Student B:

- Student A
- Student C

No empty spaces should remain.

Screenshot:



🚩 Flag 6 — Edit Trigger

Objective: Prepare your app for editing student information.

In the next CTF you'll build a full Edit Screen.

For now, create an Edit trigger.

Research simple ways Flutter can display temporary editing interfaces.

Examples include:

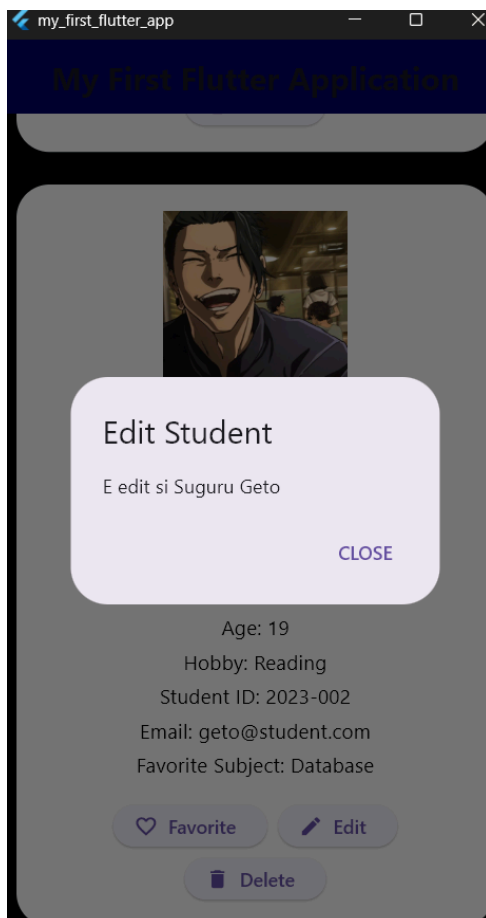
- `AlertDialog`
- `SnackBar`
- Bottom Sheet

The goal is simply to launch an edit action, not build the complete editor yet.

Test

Pressing Edit should open your chosen interface.

Screenshot:



```
onPressed: () {  
  showDialog(  
    context: context,
```

```
return AlertDialog(  
  title: const Text(  
    "Edit Student",
```

```
content: Text(  
  "E edit si ${profile.name}",  
), // Text
```

🚩 Flag 7 — Multiple Actions, Independent State

Objective: Make every Student Card manage interactions correctly.

This is your final interaction challenge.

Verify that every student behaves independently.

Example scenario:

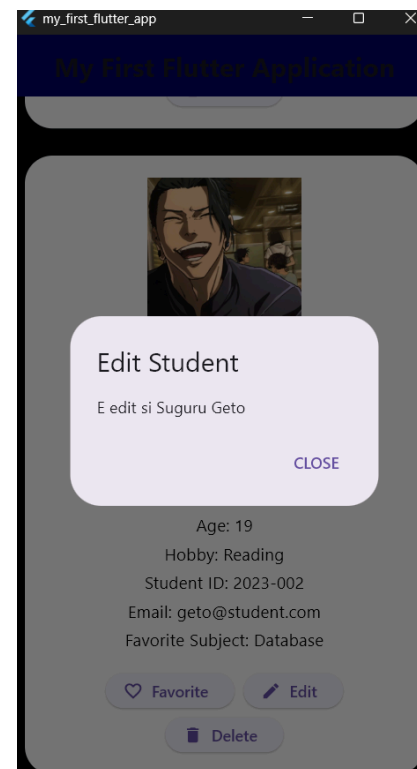
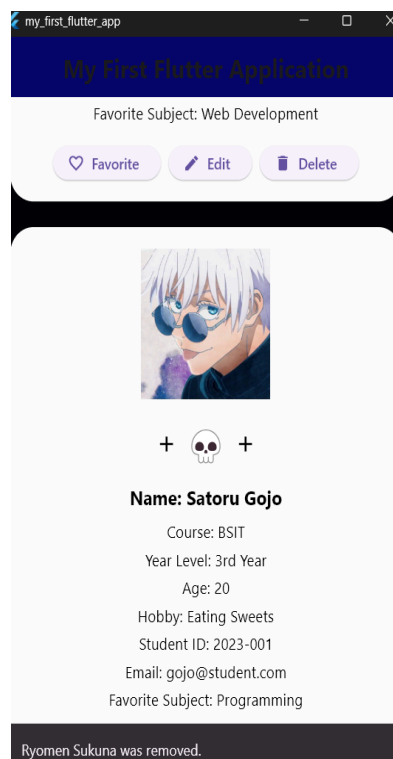
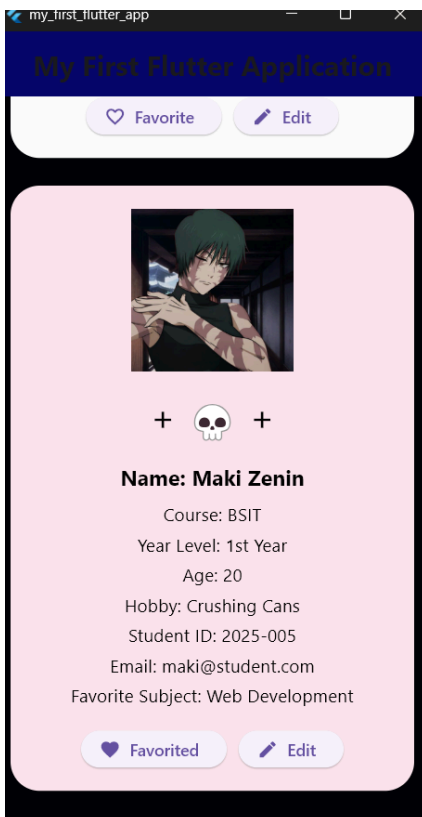
- Favorite Student A.
- Delete Student C.
- Open Edit for Student B.

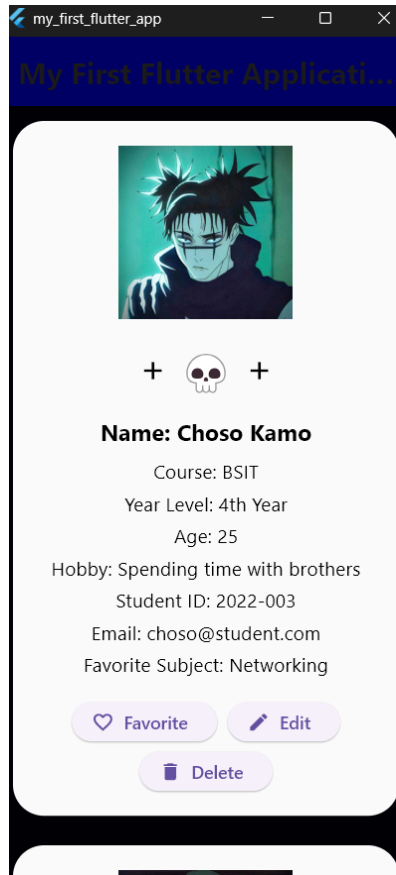
Your application should continue functioning correctly without affecting unrelated students.

Test Checklist

- Favorite works independently.
- Delete updates the list immediately.
- Edit still opens correctly.
- Remaining students keep their own states.

Screenshot:





🚩 Flag 8 — Save Your Progress

Objective: Commit and push your completed Interactive Student Directory.

Use a proper Git commit message.

Branch:

`feature/student-interactions`

Screenshot:

- Successful commit
- Current branch
- Terminal showing the commit

```
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$ git branch
  feature/data-ui
  feature/exam
  feature/exam_2
* feature/student-interactions
  feature/student-list
  feature/widgets
  main

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-interactions)
$ git status
On branch feature/student-interactions
Your branch is up to date with 'origin/feature/student-interactions'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   FlutterActivities/my_first_flutter_app/lib/main.dart

no changes added to commit (use "git add" and/or "git commit -a")

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-interactions)
$ git add .

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-interactions)
$ git commit -m "feat: submit CTF 5"
[feature/student-interactions acb6a91] feat: submit CTF 5
1 file changed, 161 insertions(+), 75 deletions(-)

WINDOWS 11@DESKTOP-CAMLQK1 MINGW64 /c/IT314AB_Caballero (feature/student-interactions)
$ git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 1.78 KiB | 1.78 MiB/s, done.
Total 6 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/mykyliocaballero/IT314AB_Caballero.git
   203f268..acb6a91  feature/student-interactions -> feature/student-interactions
```